

PIYUSH AGARWAL

Rupnagar, Punjab

☎ 8979560165 ✉ piyushaga2005@gmail.com [in LinkedIn](#) [e Portfolio](#) [g GitHub](#)

Experience

[Practers](#) | *Co-Founder & Chief Technology Officer*

Apr 2026 - Present

- Lead engineering and **0→1 architecture** for an AI interview-prep and hiring platform (**420+ users**), owning the full Next.js, Fastify, and PostgreSQL stack across web, real-time, and data layers.
- Built resume-aware AI mock interviews on Groq voice/text over authenticated WebSockets, delivering rubric-scored reports at **sub-1s** voice latency across **1,000+ interview sessions**.
- Shipped secure coding, contest, and proctored-assessment modules on Judge0 and Redis/BullMQ, evaluating **5,000+ submissions** against hidden tests with real-time verdicts and ATS resume analysis.

Swyclr Ventures | *SDE Intern (Gemini Funded Startup)* | [Certificate](#)

Jan 2026 - Jun 2026

- Engineered a **real-time barter-matching engine** using a double opt-in (mutual-interest) algorithm over Supabase/PostgreSQL with RLS, driving swipe-based discovery, instant match creation, and Expo push notifications for a cross-platform React Native marketplace.
- Built a **daily live-auction and escrow-backed deal system** with time-windowed, **concurrency-safe bidding**, monotonic bid validation, and real-time leader ranking, holding funds in **escrow** until swap completion to guarantee **transactional integrity** across simultaneous bidders, integrated with Razorpay/RevenueCat payment flows.

Education

The LNM Institute of Information Technology

2024 - Expected 2028

B.Tech in Communication and Computer Engineering | CGPA: 7.52

Jaipur, Rajasthan

Skills

Languages: C++ , Python , JavaScript , TypeScript , SQL , C

Frameworks & Technologies: React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, React Native, WebSockets

Databases & Infrastructure: PostgreSQL, MongoDB, Supabase, Redis, Prisma, Docker, Git, Judge0, Cloudflare R2

AI/ML & Coursework: LLMs, System Design, DSA, OOP, DBMS, OS, CN

Coding Profiles

LeetCode (450+ Questions)(Rating : 1872): leetcode.com/piyushagarwal-55

Codeforces (220+ Questions): codeforces.com/profile/piyushagarwal-5

GeeksForGeeks (200+ Questions): [geeksforgeeks.org/user/piyushagarwal-55](https://www.geeksforgeeks.org/user/piyushagarwal-55)

Projects

[Sheet AI Pro](#) | *Next.js, Node.js, WebSockets, PostgreSQL*

[Live Demo](#)

- Architected a collaborative spreadsheet using **Operational Transformation (OT)** over WebSockets, supporting **20+ concurrent editors** with conflict free sync and reconnect safe state.
- Engineered an **AST + DAG formula engine** for dependency resolution, achieving **sub-100ms** recalculation across **2,500+ cells** via incremental recomputation with reconnect-safe state management.

[SchemaForge - Database Migration Tool](#) | *TypeScript, PostgreSQL, Next.js*

[Live Demo](#)

- Designed a **compiler style engine** (Oracle DDL → database agnostic **IR** → PostgreSQL) that **topologically sorts** the foreign key graph via **Kahn's algorithm** with **cycle detection** to guarantee a valid table creation order.
- Built a **precision aware type resolver** preventing silent **integer overflow** (**NUMBER(10,0)** → **BIGINT**) and applied each migration in one **ACID transaction** with atomic **rollback** + dry run, cutting errors from **15-20%** to **<1%**.

[ShopSage](#) | *Next.js, FastAPI, MongoDB, AWS Bedrock, Llama 3*

[Live Demo](#)

- Built an AI shopping assistant on **AWS Bedrock (Llama 3-70B)** with multi-agent workflows for search, recommendations, and budget aware optimization across a **1,000+ product** catalog.
- Implemented **generative UI** and virtual try-on with **MongoDB-backed** product, cart, and session flows over FastAPI, enabling intent based discovery from natural language prompts at **sub-2s** response time.

Academic/Extra Curricular Achievements

- **Top Code Contributor at Practers** ([Proof](#)) – Authored the **highest line count on the team** (**210,000+ additions** across **105 commits**) on a production, multi-service platform, peaking at **92,000+ additions** in a single week.
- **Leetcode Knight(Top 5% Globally)** - Demonstrating strong DSA, problem-solving, and competitive programming skills.
- **Top 5% Among All Participants** – Google Big Code Hackathon 2026, by clearing **3 competitive coding rounds**, highlighting competitive programming proficiency.
- **AWS Hackathon** ([Certificate](#)) – Ranked among the **Top 1% of all participants**, shortlisted for an **AI/ML project**.
- **Top 10 Team** ([Certificate](#)) – The Unstoppable Hackathon for **Stability Nexus**, with Solidity contracts.
- **Teaching Assistant** ([Certificate](#)) – Served as a **Teaching Assistant** for **two consecutive semesters**, mentoring students during labs and resolving academic queries.